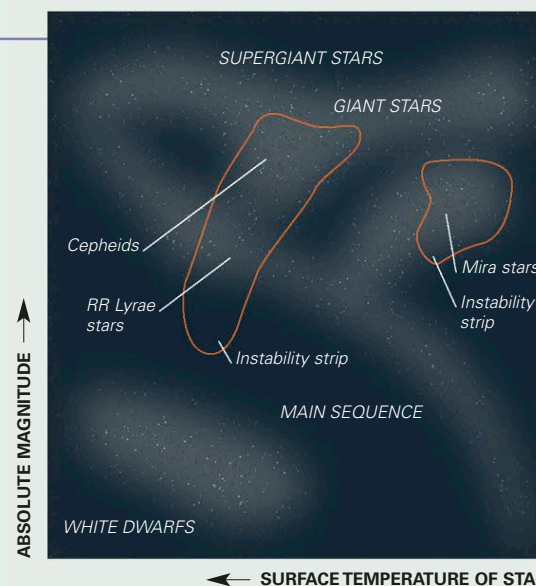


HELIUM FLASHING

Once hydrogen burning has produced a core of helium, if its temperature reaches higher than about 180 million °F (100 million °C), the helium will be fused together to form carbon. In stars of around 2 to 3 solar masses, helium burning can start in an explosive process called a helium flash. As the core collapses after hydrogen burning, it temporarily arrives at a dormant or “degenerate” state as the collapse is halted by the pressure between the helium’s electrons. The temperature continues to rise, but the dormant core does not change in pressure, so it does not expand and cool. The rising temperature causes the helium to burn at an increasing rate, causing a “flash” that rids the core of the degenerate electrons. In higher-mass stars, the temperature rises high enough for helium fusion to begin before the core becomes degenerate.

INSTABILITY STRIPS

Many red giants and supergiants are pulsating variable stars that appear in regions of the H-R diagram (right, see also p.232) called instability strips. Three types of variable star are shown here.

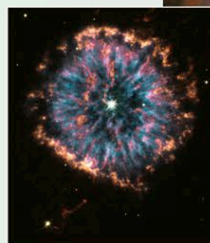


WOLF-RAYET STARS

Massive stars, of about 10 solar masses, that have strong, broad emission lines in their spectra (see p.35) but few absorption lines are named Wolf-Rayet stars, after Charles Wolf and Georges Rayet (see p.264), who discovered them in 1867. They are hot, luminous stars whose strong stellar winds have blown away their outer atmospheres, revealing the stars’ inner layers. They are broadly classified as WN, WC, and WO stars, depending on their spectra. The emission lines of WN stars are dominated by hydrogen and nitrogen, those of WC stars by carbon and helium, and those of WO by oxygen as well as carbon and helium. More than half of the known Wolf-Rayet stars are members of binary systems (see pp.274–275) with O or B stars as companions. It is believed that the Wolf-Rayet star was originally the more massive partner but lost its outer envelope to the companion star.

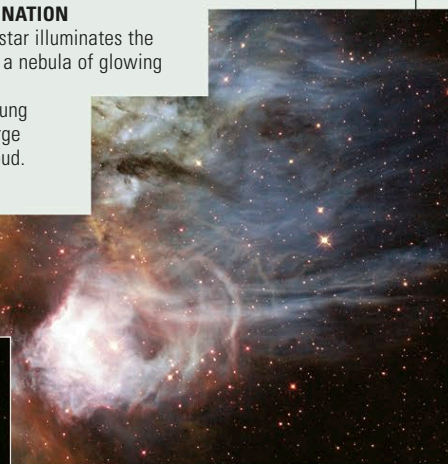
GREAT ILLUMINATION

A Wolf-Rayet star illuminates the heart of N44C, a nebula of glowing hydrogen gas surrounding young stars in the Large Magellanic Cloud.



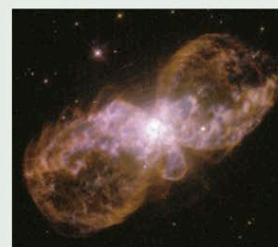
STRONG WINDS

The planetary nebula NGC 6751 may have a Wolf-Rayet star at its center. Its strong winds created the elaborate filaments.



PLANETARY NEBULAE

Planetary nebulae are heated halos of material shed by dying stars. They were termed planetary nebulae by William Herschel in 1785 because of their disklike appearance through 18th-century telescopes. Planetary nebulae include some of the most stunning sights in the universe, contorted into various shapes by magnetic fields and the orbital motion of binary systems (see pp.274–275). They are composed of low-density gas thrown off by low-mass stars in the red-giant phase of their lives, and this gas is heated by the ultraviolet radiation given off by the hot inner cores of the dying stars. This stage of a star’s life is relatively short. Eventually, the planetary nebula will disperse back into the interstellar medium, enriching the material there with the elements that have been produced by its parent star. These elements include hydrogen, nitrogen, and oxygen. At one time, the oxygen identified in the emission spectra of planetary nebulae (see p.35) was regarded as a new element called nebulium. It was later realized that “forbidden” emission lines of oxygen were present—because under usual conditions on Earth, they are very unlikely to occur. The central stars of planetary nebulae are among the hottest stars known. They are the contracting cores of red giants evolving into white dwarfs. Some planetary nebulae have been observed surrounding the resulting white dwarf. Current studies of planetary nebulae are revealing new facts about the late evolution of red giants and the manner of mass loss from these aging stars.

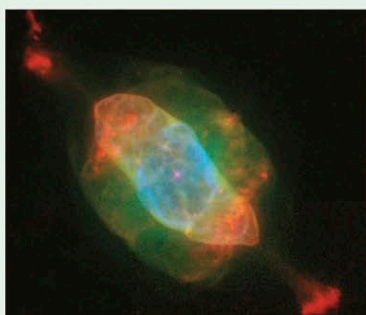


BUTTERFLY NEBULA

The Hubble 5 nebula is a prime example of a “butterfly” or bi-polar nebula, created by the funneling of expanding gas.

RING-SHAPED NEBULA

The dim star at the center of this image has produced the ring-shaped nebula around it. The nebula (NGC 3132) is crossed by dust lanes and surrounded by a cooler gas shell.



UNUSUAL NEBULA

The Saturn Nebula was shaped by early ejected material confining subsequent stellar winds into jets.

late evolution of red giants and the manner of mass loss from these aging stars.